

YvO Proteomics Pipeline - Stage 02: Imputation detail

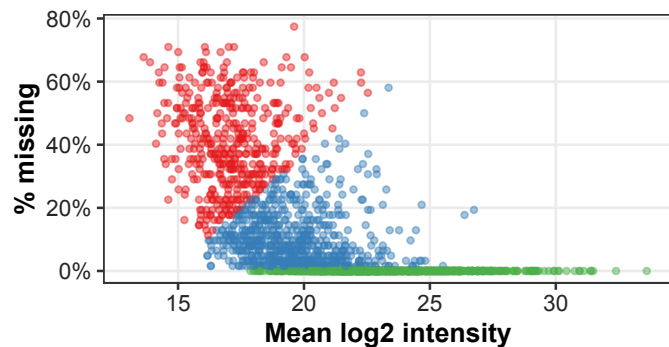
Missingness x class | method benchmark | MNAR audit | sample integrity | 2026-04-16

A

Per-protein missingness x class

2,113 proteins | *k*-means on mean intensity

● Complete: 962 (45.5%) ● MAR: 705 (33.4%) ● MNAR: 446 (21.1%)

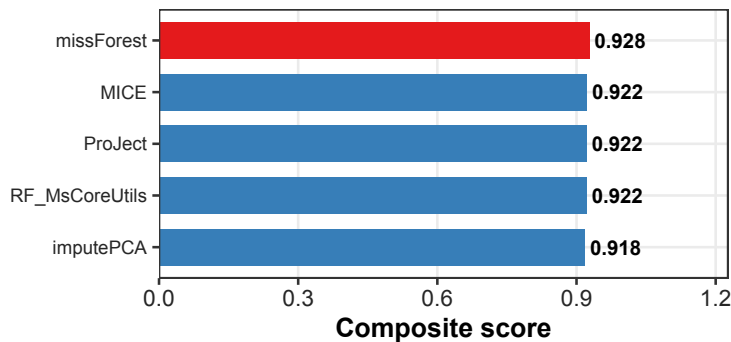


B

23-method benchmark (top 5)

Composite score | winner: missForest (0.928)

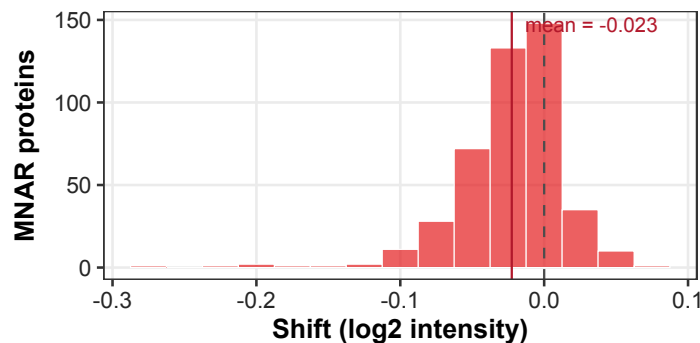
■ Top 5 ■ Winner



C

MNAR audit: imputation shift

446 MNAR proteins | pre-mean vs post-mean (log2)



D

Mean intensity: observed vs imputed

$R^2 = 0.438$ | OOB = 0.109

● Old_Post ● Old_Pre ● Young_Post ● Young_Pre

